STAR-CD: Style-Aligned Remote Sensing Change Detection With Appearance-Relation Modeling

Qian Hu, Yuxuan Wang, Yan Zhang, Wen Yang, *Senior Member, IEEE*, Fang Xu, and Gui-Song Xia, *Senior Member, IEEE*

***Abstract*—Remote sensing change detection (CD) plays a crucial part in monitoring the evolution of land cover. Despite remarkable progress in recent years, existing CD models still struggle with two major challenges that lead to inaccurate change predictions. For one, current models lack sufficient global relation modeling, resulting in coarse boundary prediction and false identification of unwanted variations. For another, bitemporal images often display divergent imaging styles due to lighting, sensor, or seasonal differences, and this cross-temporal style inconsistency would amplify the pseudo-changes caused by nonsemantic variations. In this article, we present the style-alignment and appearance-relation modeling CD model (STAR-CD) to tackle the above challenges. First, we propose an appearance-relation modeling block (ARMB) to jointly explore semantic differences from both local textures and global structures of remote sensing images, effectively reducing false alarms and refining change boundary predictions. Second, we design a Fourier-inspired style alignment module (FSAM), which suppresses style-induced noise and reveals true changes by texture-style decoupling in the frequency domain. Extensive experiments on the WHU-CD, LEVIR-CD, and CDD datasets demonstrate the superiority of our STAR-CD over state-of-the-art methods. In particular, STAR-CD surpasses previous methods by 2.40% and 4.34% in $F1$ and intersection over union (IoU) on the CDD datasets, respectively. Visual comparisons further validate its strength in predicting precise change boundaries and robustly suppressing pseudo-changes.**

***Index Terms*—Appearance-relation modeling, change detection (CD), Fourier transform, remote sensing.**

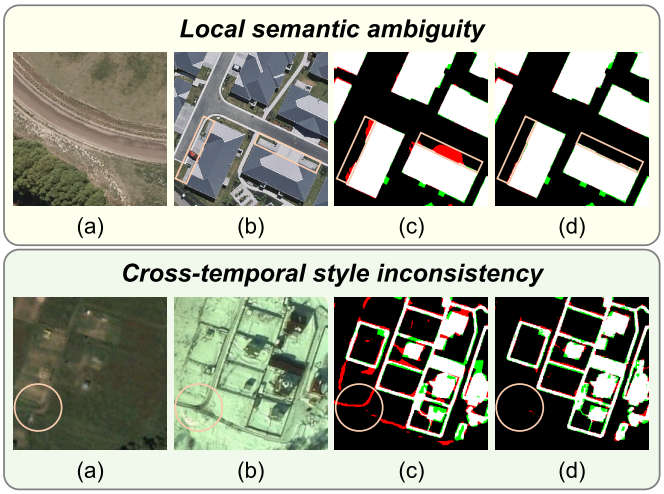


Fig. 1. Two performance bottlenecks in the existing CD model. 1) Local semantic ambiguity: models often misidentify visually similar but irrelevant changes (building-like ground in the first scene) due to insufficient modeling of global relation cues. 2) Cross-temporal style inconsistency: objects with identical semantics (roads in the second scene) may appear drastically different across time, leading to style-induced pseudo-change. (a)–(d) Bitemporal images, predictions from state-of-the-art methods, and from our STAR-CD. The true positives, true negatives, false positives, and false negatives are represented by white, black, red, and green, respectively.

I. Introduction

REMOTE sensing change detection (CD) aims to identify important and meaningful land cover variations between aerial images from different temporal periods, which serves as a crucial technique for monitoring dynamic Earth processes [1], [2]. With the exponential growth of remote sensing data, CD has become indispensable for applications like urban expansion analysis [3], [4], disaster damage assessment [5], [6], and infrastructure planning [7]. However, as visualized in Fig. 1, precisely identifying real changes and delineating their boundaries with pixel-level accuracy remains fundamentally challenging due to two inherent characteristics of bitemporal remote sensing data. First, remote sensing imagery inherently suffers from local semantic ambiguity, as small patches often lack sufficient context to distinguish between visually similar classes. For example, rooftops and concrete roads may appear similar in isolation, but correspond to different land covers. Such local ambiguities are especially problematic near change boundaries, necessitating the integration of global contextual information to supplement scene-level semantics for accurate distinction. Second, due to varying acquisition conditions, bitemporal remote sensing images often exhibit noticeable differences in tone and contrast. This cross-temporal style inconsistency can cause identical ground objects to appear markedly different across acquisitions. As a result, models may misinterpret these nonsemantic variations as real changes, thereby impairing CD performance. These two challenges underscore the necessity of strengthening global

Received 23 June 2025; revised 9 September 2025 and 10 November 2025; accepted 9 December 2025. Date of publication 12 December 2025; date of current version 22 December 2025. This work was supported in part by the National Natural Science Foundation of China (NSFC) Regional Innovation and Development Joint Fund under Grant U22A2010 and in part by NSFC under Grant 62271355. *(Corresponding author: Wen Yang.)*

Qian Hu, Yuxuan Wang, and Wen Yang are with the Electronic Information School, Wuhan University, Wuhan 430072, China (e-mail: huq1an@whu.edu.cn; wyx2017@whu.edu.cn; yangwen@whu.edu.cn).

Yan Zhang is with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: zhangyan@whu.edu.cn).

Fang Xu and Gui-Song Xia are with the School of Artificial Intelligence, Wuhan University, Wuhan 430072, China (e-mail: xufang@whu.edu.cn; guisong.xia@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3643487

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

relation modeling and suppressing style-induced noise within CD frameworks.

Before the rise of deep learning, traditional CD methods relied on shallow statistical operations, such as image differencing, image ratio, principal component analysis (PCA) [8], and change vector analysis (CVA) [9] to detect changes by measuring pixel value variations. Although these traditional methods are computationally efficient and simple, they often struggle to handle high-resolution remote sensing images, resulting in coarse change prediction. There are some other traditional methods that try to incorporate hand-crafted priors, such as morphological filters [10], [11], [12]. However, these methods typically exhibit poor generalization capabilities and require manual parameter tuning. In recent years, deep learning-based end-to-end CD methods have become mainstream [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. Early representative methods were primarily built on convolutional neural networks (CNNs) [14], [15], [16], [17], [21], [23], where bitemporal images are stacked together for feature extraction. However, researchers gradually noticed that this architecture could result in insufficient information extraction, and then proposed the widely adopted Siamese architecture. More recently, transformer-based CD methods [18], [19], [20], [24], [25], [26] have been introduced to better capture long-range dependencies and global context, achieving further gains in detection accuracy. Despite these advances, the aforementioned two fundamental challenges, the local semantic ambiguity and the cross-temporal style inconsistency, still remain major performance bottlenecks for existing CD models.

In this article, we propose a style-alignment and appearance-relation modeling CD model (STAR-CD) to tackle these limitations. In detail, our STAR-CD follows the well-adopted Siamese network architecture, with two specifically designed bitemporal feature interaction modules for solving existing problems. First, to overcome the local semantic ambiguity of remote sensing images, we design an appearance-relation modeling block (ARMB). The ARMB employs dual complementary pathways; one focuses on modeling local appearance differences through localized attention, while the other incorporates a spatial shuffle operation to enhance perception of global relational changes, thereby enabling the model to capture both local and contextual change cues. Second, to alleviate style-induced change noise, we propose a Fourier-inspired style alignment module (FSAM), where the style and structural information of the bitemporal feature are decoupled in the frequency space. Specifically, the style characteristics of bitemporal features are encoded in their respective amplitude spectra. By aligning these amplitude components, we effectively mitigate style discrepancies that could mislead the detection of real changes. Meanwhile, the phase spectra, which preserve the unique structural layouts of each temporal feature, are retained to ensure semantic integrity. This design enables our STAR-CD to effectively address the issue of style-semantic inconsistency. In summary, our contributions can be concluded as follows.

1) We pinpoint two overlooked limitations in existing CD models, the local semantic ambiguity and the cross-temporal style inconsistency, and propose a novel CD method, STAR-CD, for solving these performance bottlenecks.
2) To address the local semantic ambiguity in remote sensing imagery, we propose an ARMB. ARMB adopts a dual-path modeling mechanism that simultaneously captures fine-grained local appearance differences and holistic scene relation variation, effectively alleviating local feature fragmentation.
3) To mitigate pseudo-changes induced by cross-temporal style discrepancies, we develop an FSAM, which decomposes the style characteristics and the structural layouts of bitemporal features in the frequency domain. By explicitly aligning the former while preserving the latter, FSAM reduces style-induced noise without compromising the semantic integrity of change regions.
4) We conduct extensive qualitative and quantitative experiments on three widely used challenging CD datasets. The results demonstrate the superiority of our proposed STAR-CD against state-of-the-art CD methods.

The remainder of this article is structured as follows. Section II gives a comprehensive literature review of remote sensing CD and the feature interaction mechanisms in CD. Section III provides a detailed introduction to our proposed STAR-CD. Extensive quantitative and qualitative experiments are conducted and presented in Section IV. In Section V, we discussed the limitations of existing CD methods and possible future research directions. Finally, we draw a conclusion in Section VI.

II. Related Works

A. Remote Sensing CD

As a fundamental task in remote sensing interpretation, CD has long focused on developing reliable techniques to identify land cover variations. Early efforts in remote sensing CD primarily relied on handcrafted rules to measure bitemporal image differences. These traditional methods can be categorized into three types. Pixel-level methods identify change regions based on the difference of each pixel. For example, PCA is employed in [8] to project pixels into high-dimensional vectors for clustering, and independent component analysis is used in [27] to discriminate sources of noise from true changes. Object-level methods [28], [29], [30] capture semantic changes by considering region entities in temporal images. Other methods incorporate domain prior knowledge, such as morphological operations for extracting image differences [10], a low-rank prior for change probability prediction [11], and sparsity constraints for exploring the difference of multisource data [12]. Recently, DIDCD [31] introduced a difference-in-differences model, using two difference calculations to extract change intensity. Although these traditional methods are computationally efficient, they heavily rely on manual parameter tuning and exhibit limited generalization capability when applied to high-resolution and fine-grained CD scenarios.

The advent of deep learning shifted the paradigm toward data-driven CD, with pioneering CNN-based methods that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

directly stack bitemporal remote sensing images as inputs to regress change maps from end to end [13]. However, the entangled feature representation limits their ability to extract bitemporal differences. This limitation motivates the introduction of Siamese architectures [14], [15], [16], [17], [21], [23], which independently encode bitemporal features for explicit differential analysis. Based on this framework, several representative works are proposed. SNUNet [15] constructs a densely connected Siamese network to alleviate the loss of localization information. DMINet [16] proposes a dual-branch multilevel intertemporal network to effectively derive the change representations. Recently, SEIFNet [17] has enhanced bitemporal differences by considering the spatiotemporal characteristics and realizes multiscale complementarity through interlevel fusion. To realize precise change capture, FTANet [21] breaks away from the traditional time domain analysis paradigm and introduces frequency domain learning to enhance high- and low-frequency information.

There are several methods that adopt the Transformer frameworks, whose self-attention mechanism effectively addresses the receptive field limitations of CNNs [18], [19], [20], [24], [25], [26]. As the first Transformer-based CD method, SwinSUNet [25] proposes a pure transformer network with a Siamese U-shaped structure. BIT [24] then expresses the input images as a few visual words, which benefits the identification of the change of interest. To fully utilize the advantage of CNN and transformer, TransUNetCD [26] designs a hybrid framework and realizes low-level detail preservation. Recently, MDIPNet [20] recognizes the high computational consumption brought by the Transformer architecture and achieves lightweight CD by fully exploiting information in the multiscale feature space. While the aforementioned approaches focus on architectural innovation within discriminative models, the generative framework has also been introduced. For instance, RS-NormGAN [32] employs a generative neural network to model the complex, nonlinear radiometric transformations between multitemporal images for minimizing pseudo-changes.

Despite the significant progress made by deep learning-based CD methods, current solutions still suffer from two key limitations: local semantic ambiguity and cross-temporal style inconsistency. These challenges have not been sufficiently addressed in existing studies and motivate us to develop a new CD framework that explicitly tackles them in a unified manner.

B. Feature Interaction Strategies in CD

After the widespread adoption of Siamese network architectures in bitemporal remote sensing CD, researchers have shifted their focus toward designing effective bitemporal feature interaction strategies to enhance discriminative change representation or align feature distribution across time [14], [15], [16], [20], [33], [34], [35]. Early feature interaction strategies [14], [15], [34] primarily employed basic operations such as absolute difference or concatenation for implicit difference extraction, which rely on simple operations. Although these strategies are straightforward, they often fail to capture discriminative change patterns, especially in complex scenes.

To enhance the model's capability in extracting bitemporal discrepancy, STANet [33] designs a spatial–temporal attention module to capture discriminative cues from both spatial and temporal domains. DMINet [16] proposes an intertemporal joint-attention block to suppress CD-irrelevant interferences. MDIPNet [20] designs a cross-space perception module to establish a connection between differential feature spaces, therefore enriching semantic information. SEIFNet [17] further proposes a spatiotemporal difference enhancement module to obtain target-level global change-related information. In parallel, some other methods focus on aligning the bitemporal feature distribution [19], [35], [36], [37]. For instance, Changer [35] introduces a parameter-free feature exchange operation to reduce discrepancies. The flow-based alignment is introduced in [19] and [35] to mitigate the spatial registration error between bitemporal images.

While current feature interaction strategies demonstrate empirical improvements, most of them adopt heuristic architectures and fail to directly address the core challenges of bitemporal CD, such as local semantic ambiguity and cross-temporal style inconsistency. In contrast, our proposed STAR-CD framework incorporates two dedicated bitemporal feature interaction modules that are specifically designed to tackle these limitations: the ARMB for resolving semantic ambiguity, and the FSAM for suppressing style-induced pseudo-changes.

C. Relation Modeling in Remote Sensing

In remote sensing image processing, relation modeling has emerged as an effective means to enhance contextual understanding and semantic coherence. Such techniques have been explored in tasks like semantic segmentation [38], [39], [40], [41] and object recognition [42], [43], [44], where intraimage relations help strengthen category consistency. For example, attention-based [38] or graph-based [43] relation reasoning modules have been introduced to encode spatial interactions within a single remote sensing image, achieving improved feature discrimination under stable imaging conditions.

Nevertheless, unlike in semantic segmentation, where relation modeling aims to understand the static structural layout within a single scene, relation modeling in CD serves a quite different purpose. It not only needs to comprehend the contextual organization within each temporal image but also to determine whether the relations among objects themselves have changed. Such relational shifts are crucial for identifying real changes and suppressing pseudo ones. This makes cross-temporal relation modeling substantially more challenging, as the model must establish reliable relational correspondence between bitemporal remote sensing images that often exhibit significant differences. In response, our ARMB addresses this by explicitly modeling cross-temporal relational dynamics. Through a dual-branch attention mechanism, ARMB jointly captures local appearance changes and global relational shifts, enabling it to effectively distinguish real land-cover changes.

D. Style Alignment in Remote Sensing CD

In remote sensing CD, due to factors such as illumination variations, seasonal differences, and sensor discrepancies, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

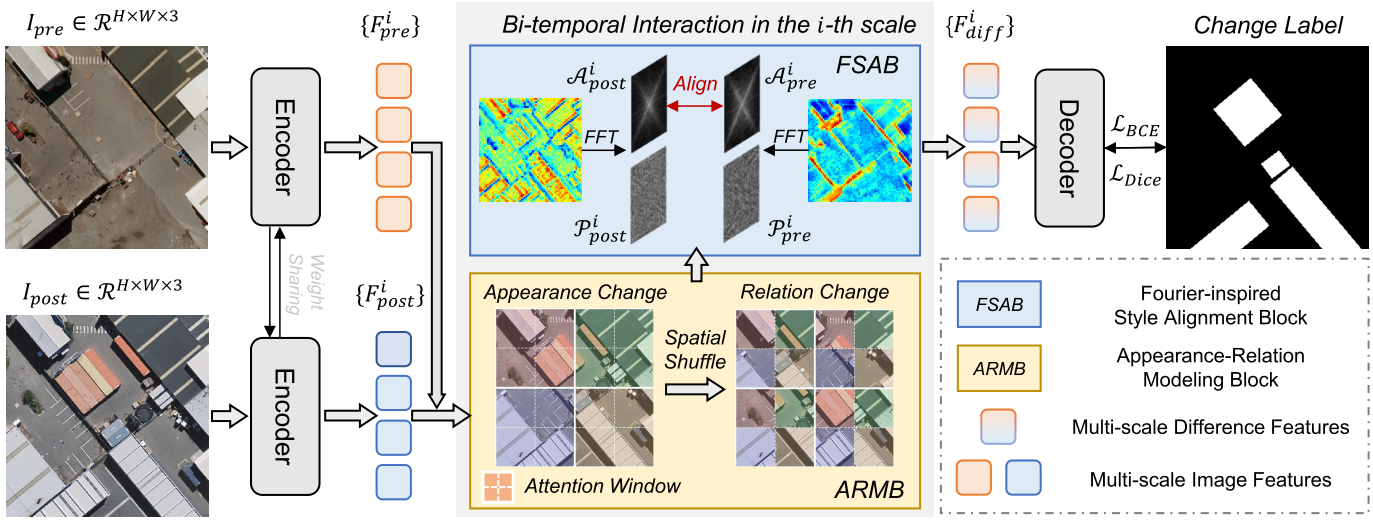


Fig. 2. Overall framework of the proposed STAR-CD. We first extract the multiscale features of the bitemporal images using a weight-shared image encoder. Then, we feed these features into ARMB for appearance-relation-aware change modeling. After that, we introduce the FSAM to align the style information of bitemporal features. Finally, the multiscale difference features are projected to the final change prediction map through a decoder.

bitemporal remote sensing images often exhibit noticeable visual discrepancies, which can collectively be referred to as style inconsistency. This inconsistency can cause objects with consistent semantics but different visual patterns to be mistakenly identified as changes. To address this issue, several studies in recent years have introduced style alignment strategies to narrow temporal discrepancies, either at the image level or the feature level. The former performs cross-temporal style transfer to make the bitemporal images visually consistent before comparison [45], [46], [47]. For example, the CycleGAN framework is introduced in [47] to achieve optical-to-SAR image translation, effectively reducing modality-induced visual discrepancies. However, these approaches usually rely on additional generative models, introducing considerable computational overhead and potential instability in the generated results. The latter attempts to align the distribution of bitemporal features through domain adaptation or feature normalization [48], [49], [50]. Representative work, such as SAFDNet [49], designs a histogram equalization-based module to normalize feature distributions of bitemporal images. However, most of these methods operate in the spatial domain, directly enforcing similarity across entire feature maps without disentangling style information from structural semantics, which may lead to suboptimal alignment.

Considering these limitations, we propose an FSAM that performs amplitude-based style alignment in the frequency space. The design of FSAM is motivated by the spectral decomposition property of the Fourier Transform, where the amplitude spectrum primarily encodes style-related global appearance, while the phase spectrum predominantly preserves the semantic structural layout. By leveraging this property, FSAM aligns the style components across different temporal domains while preserving their distinct structural semantics. Our proposed FSAM enables the model to build style-invariant temporal representations, thereby suppressing pseudo changes and enhancing the robustness of CD.

III. Methods

In this section, we will give a detailed description of the proposed remote sensing CD method, STAR-CD. As shown in Fig. 2, STAR-CD follows the common encoder-decoder structure, with two key modules, ARMB for modeling bitemporal image change from their local appearance and nonlocal relation simultaneously, and FSAM for mitigating the change ambiguity caused by style discrepancies between the bitemporal images while preserving their unique texture information. In the following, we will first introduce the overall framework of STAR-CD, followed by an in-depth discussion of ARMB and FSAM.

A. Overall Framework

The overall framework of STAR-CD is illustrated in Fig. 2. Given the bitemporal remote sensing images $I_{\text{pre}} \in \mathbb{R}^{H\times W\times 3}$ and $I_{\text{post}} \in \mathbb{R}^{H\times W\times 3}$ with spatial size $H \times W$, STAR-CD first adopts a weight-shared image encoder H_{enc} to extract the multiscale bitemporal image features, which can be formulated as

$$\left\{F^i_{\text{pre}} | i = 1, 2, 3, 4\right\} = H_{\text{enc}}\left(I_{\text{pre}}\right) \tag{1}$$

$$\left\{F^i_{\text{post}} | i = 1, 2, 3, 4\right\} = H_{\text{enc}}\left(I_{\text{post}}\right) \tag{2}$$

where F^i_{pre} and F^i_{post} are bitemporal image features at the ith scale. Then, STAR-CD models bitemporal changes by simultaneously capturing both local appearance variations and global relational differences through the proposed ARMB. For the bitemporal features at the ith scale, the above process can be formulated as

$$\hat{F}^i_{\text{pre}}, \hat{F}^i_{\text{post}} = H^i_{\text{ARMB}}\left(F^i_{\text{pre}}, F^i_{\text{post}}\right) \tag{3}$$

where H^i_{ARMB} refers to the ARMB at the ith scale. After that, the bitemporal features are projected onto the Fourier space by the proposed FSAM, where their style discrepancies are mitigated while texture information is preserved

$$F^i_{\text{diff}} = H^i_{\text{FSAM}}\left(\hat{F}^i_{\text{pre}}, \hat{F}^i_{\text{post}}\right) \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

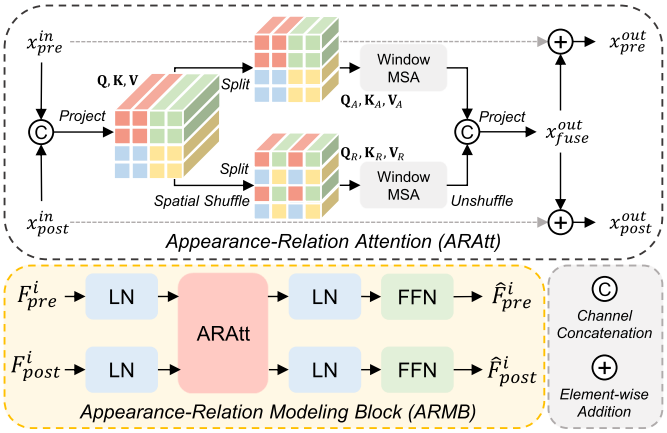


Fig. 3. Architecture of the ARMB and its key component ARAtt.

where H^i_{FSAM} refers to the FSAM at the ith scale and F^i_{diff} refers to the difference features at the ith scale. Finally, the difference features from all scales are fused and mapped to the final change prediction map $I_{\text{pred}} \in \mathbb{R}^{H\times W}$ through a decoder H_{dec}

$$I_{\text{pred}} = H_{\text{dec}}\left(\left\{F^i_{\text{diff}} | i = 1, 2, 3, 4\right\}\right). \tag{5}$$

B. Appearance-Relation Modeling Block

To achieve accurate remote sensing CD, the model is required to not only identify whether a real change has occurred, but also precisely verify whether the changed region belongs to the specific target category, even in single-category CD scenarios. However, due to the inherent local semantic ambiguity of remote sensing images, relying solely on local appearance information often fails to provide sufficient semantic guidance to accurately classify the changed regions. This limitation can result in pseudo-change or missed detections, posing a major challenge to existing CD models. To tackle this challenge, we propose the ARMB, which simultaneously models bitemporal changes from local appearance variations and global relational differences.

The architecture of the proposed ARMB is shown in Fig. 3, which consists of four layer normalizations (LNs), two feed-forward networks (FFNs), and one appearance-relation attention (ARAtt). Given the bitemporal features at the ith feature scale F^i_{pre} and F^i_{post}, the computation process of ARMB can be formulated as follows:

$$\tilde{F}^i_{\text{post}} = \text{ARAtt}\left(\text{LN}\left(F^i_{\text{post}}\right), \quad \text{LN}\left(F^i_{\text{pre}}\right)\right) + F^i_{\text{post}} \tag{6}$$

$$\hat{F}^i_{\text{post}} = \text{FFN}\left(\text{LN}\left(\tilde{F}^i_{\text{post}}\right)\right) + \tilde{F}^i_{\text{post}} \tag{7}$$

and the computation process for F^i_{pre} follows a symmetric formulation, where the residual connection is applied to F^i_{pre} accordingly.

The ARAtt is responsible for modeling both local appearance changes and global relational changes in bitemporal remote sensing images, which is inspired by the success of the shuffle operation in vision transformers [51], [52], [53]. Specifically, denoting the input bitemporal features as $x^{\text{in}}_{\text{pre}} \in \mathbb{R}^{H'\times W'\times C'}$ and $x^{\text{in}}_{\text{post}} \in \mathbb{R}^{H'\times W'\times C'}$, these features are stacked and projected to get the query, key and value matrix $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{H'\times W'\times 2C'}$

$$\begin{aligned}\mathbf{Q} &= \mathbf{W}_Q\left(x^{\text{in}}_{\text{pre}}, x^{\text{in}}_{\text{post}}\right), \quad \mathbf{K} = \mathbf{W}_K\left(x^{\text{in}}_{\text{pre}}, x^{\text{in}}_{\text{post}}\right)\\ \mathbf{V} &= \mathbf{W}_V\left(x^{\text{in}}_{\text{pre}}, x^{\text{in}}_{\text{post}}\right)\end{aligned} \tag{8}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{2C'\times 2C'}$ are learnable projection weights. Then, these queries, keys, and values are equally split along the channel dimension into two parts for appearance modeling and relation modeling, respectively,

$$\left[\mathbf{Q}_A, \mathbf{Q}_R\right] = \mathbf{Q}, \quad \left[\mathbf{K}_A, \mathbf{K}_R\right] = \mathbf{K}, \quad \left[\mathbf{V}_A, \mathbf{V}_R\right] = \mathbf{V}. \tag{9}$$

For appearance change modeling, split features $\mathbf{Q}_A, \mathbf{K}_A, \mathbf{V}_A \in \mathbb{R}^{H'\times W'\times C'}$ are first partitioned into nonoverlapping $M \times M$ windows, resulting in features with the size of $(H' \times W')/(M^2) \times M^2 \times C$. After that, the standard window-based multihead self-attention (window MSA) is introduced to capture local appearance change. The above process can be expressed as

$$\mathbf{A}_A = \mathbf{V}_A\text{Softmax}\left(\frac{\mathbf{Q}_A\mathbf{K}_A}{\sqrt{d}} + \mathbf{P}_A\right) \tag{10}$$

where d refers to the dimension of features, $\mathbf{P}_A$ is a learnable positional embedding, and $\mathbf{A}_A$ is the output of the appearance modeling branch.

For relation change modeling, we first employ window partitioning on split features $\mathbf{Q}_R, \mathbf{K}_R, \mathbf{V}_R$. Then, we conduct a spatial shuffle operation to establish global relation dependencies. Specifically, we aggregate pixels that occupy the same relative position within different $M \times M$ windows, forming a new window of size $(H' \times W')/(M^2)$. This operation enables global spatial perception within the transformed feature of size $M^2\times(H'\times W')/(M^2)\times C'$, while maintaining the computational efficiency of window-based attention. The window MSA is also employed on the shuffled features, which can be formulated as

$$\mathbf{A}_R = \mathbf{V}_R\text{Softmax}\left(\frac{\mathbf{Q}_R\mathbf{K}_R}{\sqrt{d}} + \mathbf{P}_R\right). \tag{11}$$

Finally, the attention result $\mathbf{A}_R$ is reshuffled back to the size of $(H' \times W')/(M^2) \times M^2 \times C$ and concatenated with the appearance attention result $\mathbf{A}_A$. The fused attention features are then linearly projected to obtain $x^{\text{out}}_{\text{fuse}}$. To preserve the original feature representation, the final output features of ARAtt are obtained by adding $x^{\text{out}}_{\text{fuse}}$ to the corresponding bitemporal input features as residual connections

$$x^{\text{out}}_{\text{pre}} = x^{\text{in}}_{\text{pre}} + x^{\text{out}}_{\text{fuse}}, \quad x^{\text{out}}_{\text{post}} = x^{\text{in}}_{\text{post}} + x^{\text{out}}_{\text{fuse}}. \tag{12}$$

Equipped with the proposed ARMB, the CD model is able to overcome the inherent local semantic ambiguity of remote sensing images. By jointly modeling local appearance variations and global relational changes between bitemporal images, it effectively captures changes belonging to specific target categories, while reducing missed detections and pseudo-changes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

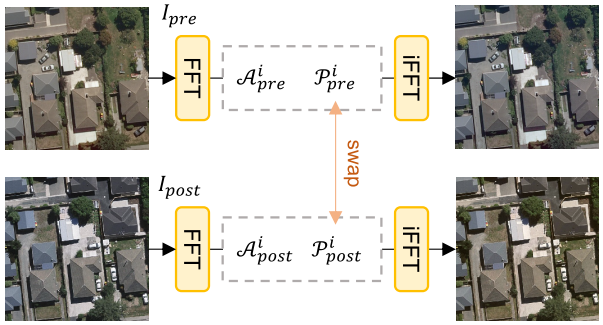


Fig. 4. Motivation of our proposed FSAM. By simply swapping the amplitude spectrum of bitemporal images while preserving their original phase components, the structural layout is retained, whereas the visual style is exchanged.

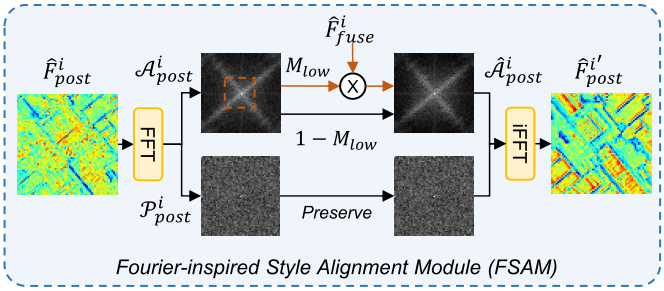


Fig. 5. Architecture of the FSAM. The FFT and iFFT refer to fast Fourier transform and inverse fast Fourier transform, respectively.

C. Fourier-Inspired Style Alignment Module

Bitemporal remote sensing images often exhibit substantial differences, caused by variations in illumination conditions, seasonal characteristics, and sensor settings. These variations are typically manifested as significant style discrepancies between the two images. Such cross-temporal style discrepancies can easily lead to misinterpretation of nonsemantic changes as true land cover changes. For example, the appearance of vegetation varies greatly between spring and fall, yet their actual semantics are the same.

To disentangle style-induced artifacts, we propose the FSAM to decouple bitemporal features in the frequency domain and align their style information. The motivation of FSAM stems from the spectral decomposition properties of the Fourier transformation for disentangling style-content information [54], [55], [56]. To be specific, the amplitude primarily encodes global appearance and style, while the phase component preserves structural layout and spatial details. As illustrated in Fig. 4, when swapping the amplitude spectrum of bitemporal images while retaining their original phase components, the resulting images retain the structural content of the source images but exhibit the style appearance of the swapped counterpart. This phenomenon motivates us to perform amplitude-based style alignment in the frequency space to suppress style-induced pseudo-changes and enhance the robustness of CD.

The framework of the proposed FSAM is depicted in Fig. 5. For simplicity, we only demonstrate the style alignment process for post-change image features $\hat{F}^i_{\text{post}}$, and the computation process for $\hat{F}^i_{\text{pre}}$ follows a symmetric formulation. The core idea of FSAM is utilizing the fusion result of the bitemporal features $\hat{F}^i_{\text{fuse}}$ as the guidance information for bitemporal style alignment. Specifically, we first obtain the fusion result $\hat{F}^i_{\text{fuse}}$ as follows:

$$\hat{F}^i_{\text{fuse}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(\hat{F}^i_{\text{pre}}, \hat{F}^i_{\text{post}}\right)\right) \tag{13}$$

where $\text{Conv}_{1\times1}$ is one convolution layer with kernel size 1×1. Then, we employ fast Fourier transform (FFT) on $\hat{F}^i_{\text{post}}$ to decompose it into the amplitude spectrum $\mathcal{A}^i_{\text{post}}$ and phase spectrum $\mathcal{P}^i_{\text{post}}$

$$\mathcal{A}^i_{\text{post}}(u,v) = \sqrt{\text{Re}\left(\mathcal{F}^i_{\text{post}}(u,v)\right)^2 + \text{Im}\left(\mathcal{F}^i_{\text{post}}(u,v)\right)^2} \tag{14}$$

$$\mathcal{P}^i_{\text{post}}(u,v) = \tan^{-1}\left(\frac{\text{Im}\left(\mathcal{F}^i_{\text{post}}(u,v)\right)}{\text{Re}\left(\mathcal{F}^i_{\text{post}}(u,v)\right)}\right) \tag{15}$$

where u and v are coordinates in the frequency domain. Re and Im indicate the real part and imaginary part of the Fourier spectrum. For the amplitude spectrum $\mathcal{A}^i_{\text{post}}$, it generally encodes the style characteristic of the feature, particularly within its low-frequency components [54], [57]. These style-related attributes can adversely influence CD accuracy by introducing domain-specific biases. Conversely, the phase spectrum preserves critical semantic structural layout information that is essential for distinguishing land cover changes. The above observation motivates us to align the style representations through amplitude spectrum modulation while maintaining the phase spectrum to preserve semantic patterns. Specifically, we introduce a low-pass mask M_{low} to preserve low-frequency components, which represents the overall style of the features. This mask retains value 1 only within a centered rectangular region spanning $\beta H \times \beta W$ pixels. After that, the amplitude modulation process for $\mathcal{A}^i_{\text{post}}$ can be formulated as

$$\hat{\mathcal{A}}^i_{\text{post}} = M_{\text{low}}\mathcal{A}^i_{\text{post}}\left(1 + \tanh\left(\hat{F}^i_{\text{fuse}}\right)\right) + (1 - M_{\text{low}})\,\mathcal{A}^i_{\text{post}} \tag{16}$$

where the tanh activation function is introduced as a numerical constraint. After amplitude modulation, the output of FSAM can be obtained by applying the inverse FFT (iFFT) to the modulated amplitude $\hat{\mathcal{A}}^i_{\text{post}}$ and the original phase spectrum $\mathcal{P}^i_{\text{post}}$

$$F^{i\prime}_{\text{post}}(x,y) = \text{iFFT}\left(\hat{\mathcal{A}}^i_{\text{post}} \cdot e^{j\mathcal{P}^i_{\text{post}}}\right). \tag{17}$$

The proposed FSAM establishes style-invariant representation across time through frequency-space amplitude modulation, effectively filtering out style-specific variations while preserving semantic structural layout, allowing the CD model to better distinguish change regions.

Through appearance–relation modeling to resolve local semantic ambiguities and style alignment to suppress cross-temporal inconsistencies, their complementary effects enable us to obtain more reliable bitemporal difference features at the ith scale

$$F^i_{\text{diff}} = \text{abs}\left(F^{i\prime}_{\text{pre}} - F^{i\prime}_{\text{post}}\right) \tag{18}$$

and the difference features from all feature scales are merged and fed into the decoder for final prediction.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

D. Loss Function

As for the loss function, following previous works [17], [21], we adopt a combination of the widely used binary cross-entropy loss L_{BCE} [58] and the dice loss L_{Dice} [59]. These loss functions can be formulated as follows:

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{19}$$

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon} \tag{20}$$

where y_i is the ground truth label of the ith pixel, $\hat{y}_i$ is prediction probability the ith pixel, N is the number of pixels, and ϵ is set to 10^{-10} to avoid division by zero. The overall loss of our STAR-CD can be expressed as

$$L_{\text{overall}} = 0.5 * L_{\text{BCE}} + 0.5 * L_{\text{Dice}}. \tag{21}$$

For simplicity and general applicability, we adopt a balanced loss weight setting, which provides stable performance across datasets.

IV. Experiments and Results

A. Implementation Details

In our proposed STAR-CD, we use the Segformer-B0 [60] as the image encoder $H_{\text{enc}}(\cdot)$ to extract the multiscale bitemporal image features. The four scales of features extracted by Segformer-B0 have 32, 64, 160, and 256 feature channels, and their spatial sizes are 1/4, 1/8, 1/16, and 1/32 of the input image size, respectively. For the image decoder $H_{\text{dec}}(\cdot)$, we adopt the implicit feature alignment function from [61] to efficiently and precisely aggregate features from different levels. The window size M in ARMB is set to 8.

For network training, the proposed STAR-CD is implemented on an NVIDIA RTX 3090 Ti. We set the batch size to 8 and use Adam Optimizer with an initial learning rate of $1e^{-4}$. The total training epochs are set to 200.

For performance evaluation and comparison, we select nine representative state-of-the-art methods, including FC-EF [13], FC-Siam-Conc [13], FC-Siam-Diff [13], SNUNet [15], BIT [24], DMINet [16], SEIFNet [17], and FTANet [21]. These methods encompass the mainstream architectures in the field of CD, covering both CNN-based and transformer-based approaches. For a fair comparison, all comparison methods are trained with the same data and following their official setting. As for the evaluation metrics, five commonly used accuracy metrics are introduced. They are precision (Pre), recall (Rec), overall accuracy (OA), $F1$ score ($F1$), and intersection over union (IoU). The definition of these metrics is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{22}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{23}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{24}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{25}$$

TABLE I

Quantitative Result on the WHU-CD Dataset. Bold Represents the Best Result, and Underlined Represents the Second Best. All Values Are Reported in Percentage (%)

Methods	Pre	Rec	OA	F1	IoU
FC-EF [13]	91.55	90.11	99.27	90.82	83.19
FC-Siam-Conc [13]	86.77	90.55	99.07	88.62	79.57
FC-Siam-Diff [13]	84.79	92.36	99.04	88.42	79.24
SNUNet [15]	86.06	90.41	99.04	88.18	78.87
BIT [24]	93.02	<u>92.85</u>	99.44	92.93	86.80
DMINet [16]	<u>95.09</u>	92.12	<u>99.49</u>	<u>93.58</u>	<u>87.93</u>
SEIFNet [17]	92.37	**92.95**	99.41	92.66	86.32
FTANet [21]	92.64	92.83	99.42	92.74	86.46
STAR-CD	**96.04**	92.45	**99.55**	**94.21**	**89.05**

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{26}$$

where the TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively.

B. Experimental Results on WHU-CD Dataset

The WHU-CD dataset [62] is for building CD, which is generated from a pair of 0.7m aerial images with a size of 32 507 × 15 354. In our implementation, we crop the original images into 256 × 256 patches. The image numbers in the training, validation, and testing datasets are 5947, 743, and 744, respectively.

The quantitative comparison results on the WHU-CD dataset are reported in Table I. As shown in the results, our STAR-CD achieves state-of-the-art performance on the WHU-CD dataset, particularly excelling in terms of $F1$ score, which reaches 94.21%, and IoU, which reaches 89.05%. These two metrics are critical indicators of comprehensive CD capability. Notably, STAR-CD surpasses the second-best method, DMINet, by 0.63% in $F1$ score and 1.12% in IoU, demonstrating balanced improvements in precision and recall through our ARMB module, which jointly analyzes local appearance and global relational changes. Although BIT and SEIFNet achieve competitive recall values, this performance comes at the cost of reduced precision. In contrast, our proposed STAR-CD achieves the highest precision while achieving comparable recall. This result can be attributed to the proposed FSAM module, which effectively mitigates style-induced pseudo-changes. In Fig. 6, we further present the visualization results on the WHU-CD dataset. Notably, in the first scene, the STAR-CD method precisely captures the changes among densely arranged buildings. Compared with other methods, the detected change maps exhibit sharper and more accurate boundaries, which can be attributed to the proposed FSAM that alleviates the interference caused by bitemporal style inconsistencies. In the third scene, STAR-CD successfully identifies a change region that all other methods fail to detect.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

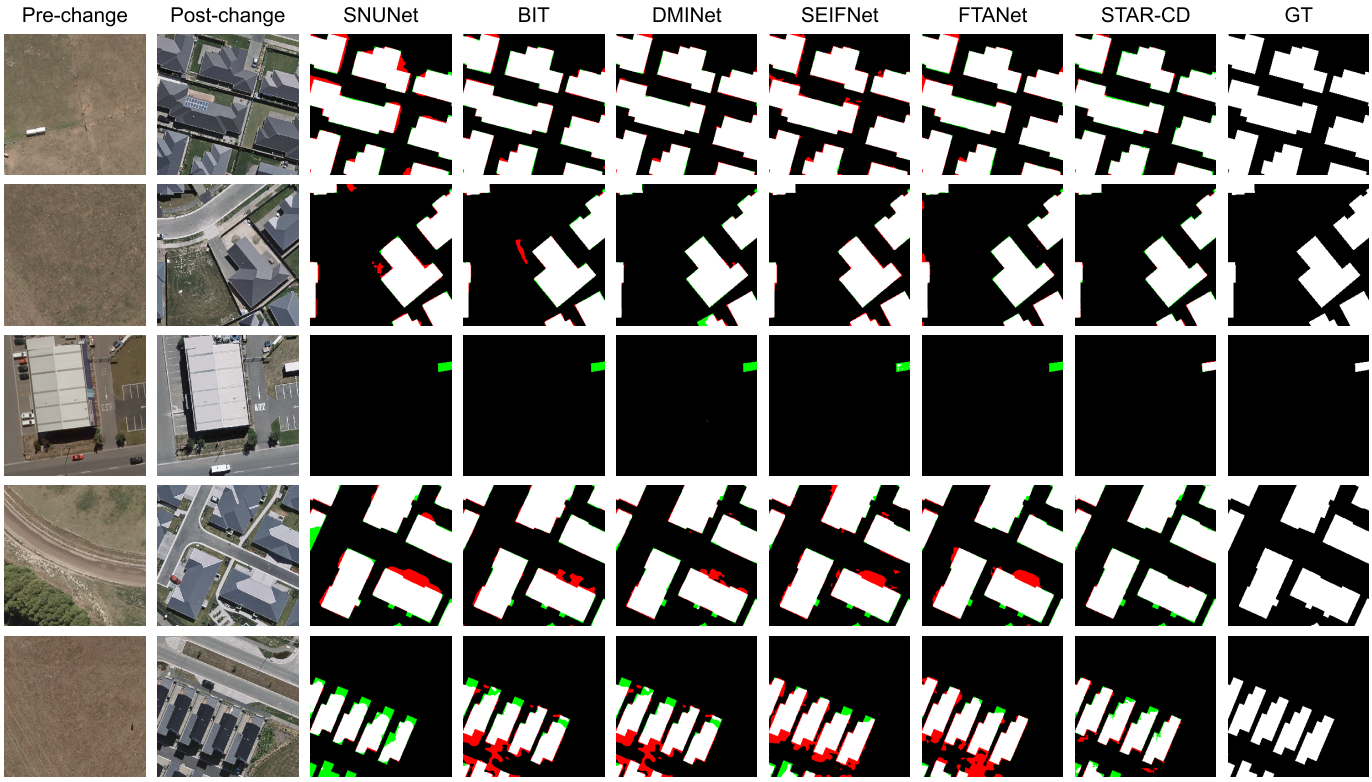


Fig. 6. Visual comparison result between our STAR-CD and the other state-of-the-art methods on the WHU-CD dataset. The true positives, true negatives, false positives, and false negatives are represented by white, black, red, and green, respectively.

In the fourth scene, other methods incorrectly recognize the regions connected to houses as part of the houses themselves, which undermines the detection accuracy. In contrast, our method accurately delineates building boundaries and localizes the actual change areas with high precision. These advantages are credited to the proposed ARMB, which captures cues from both appearance and global relations, enabling more precise change localization. The visual results further demonstrate the superiority of our method.

C. Experimental Results on LEVIR-CD Dataset

The LEVIR-CD dataset [33] is a large-scale remote sensing building CD dataset, which contains 637 very high-resolution Google Earth image patch pairs with a size of 1024×1024 pixels, and the fully annotated LEVIR-CD dataset contains 31 333 change building instances. Following the official dataset split setting, the original images are cropped into patches with a size of 256×256. The image numbers in the training, validation, and testing datasets are 7120, 1024, and 2048, respectively.

The quantitative comparison results on the LEVIR-CD dataset are presented in Table II. Our STAR-CD demonstrates consistent performance improvements across all evaluation metrics, achieving state-of-the-art results on this challenging dataset. Specifically, STAR-CD surpasses the second-best method, FTANet, by 0.41% in $F1$ score and 0.69% in IoU. While FTANet achieves competitive recall, its lower precision indicates sensitivity to pseudo-change interference caused by style differences between bitemporal images—a limitation effectively addressed by our FSAM module. Some visualization results on LEVIR-CD are further depicted in Fig. 7. As can be seen from the results, in the first scene, our proposed STAR-CD accurately recognizes two buildings with significant appearance differences as unchanged areas, while other methods generate false predictions due to style variance. This improvement stems from FSAM's effective suppression of pseudo-changes, as evidenced in the third and fourth scenes. Meanwhile, in the second scene, STAR-CD correctly captures most of the outlines of the large changed building, while the other methods generally exhibit incomplete

TABLE II
QUANTITATIVE RESULT ON THE LEVIR-CD DATASET. BOLD REPRESENTS THE BEST RESULT, AND UNDERLINED REPRESENTS THE SECOND BEST. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Methods	Pre	Rec	OA	F1	IoU
FC-EF [13]	90.84	87.46	98.91	89.11	80.37
FC-Siam-Conc [13]	91.64	88.24	98.99	89.91	81.67
FC-Siam-Diff [13]	90.91	89.11	98.99	90.00	81.82
SNUNet [15]	89.52	87.85	98.86	88.68	79.66
BIT [24]	91.30	88.83	99.00	90.04	81.89
DMINet [16]	91.39	89.26	99.02	90.30	82.33
SEIFNet [17]	<u>92.09</u>	88.25	99.01	90.13	82.03
FTANet [21]	91.62	<u>89.35</u>	<u>99.04</u>	<u>90.47</u>	<u>82.60</u>
STAR-CD	**92.21**	**89.59**	**99.08**	**90.88**	**83.29**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

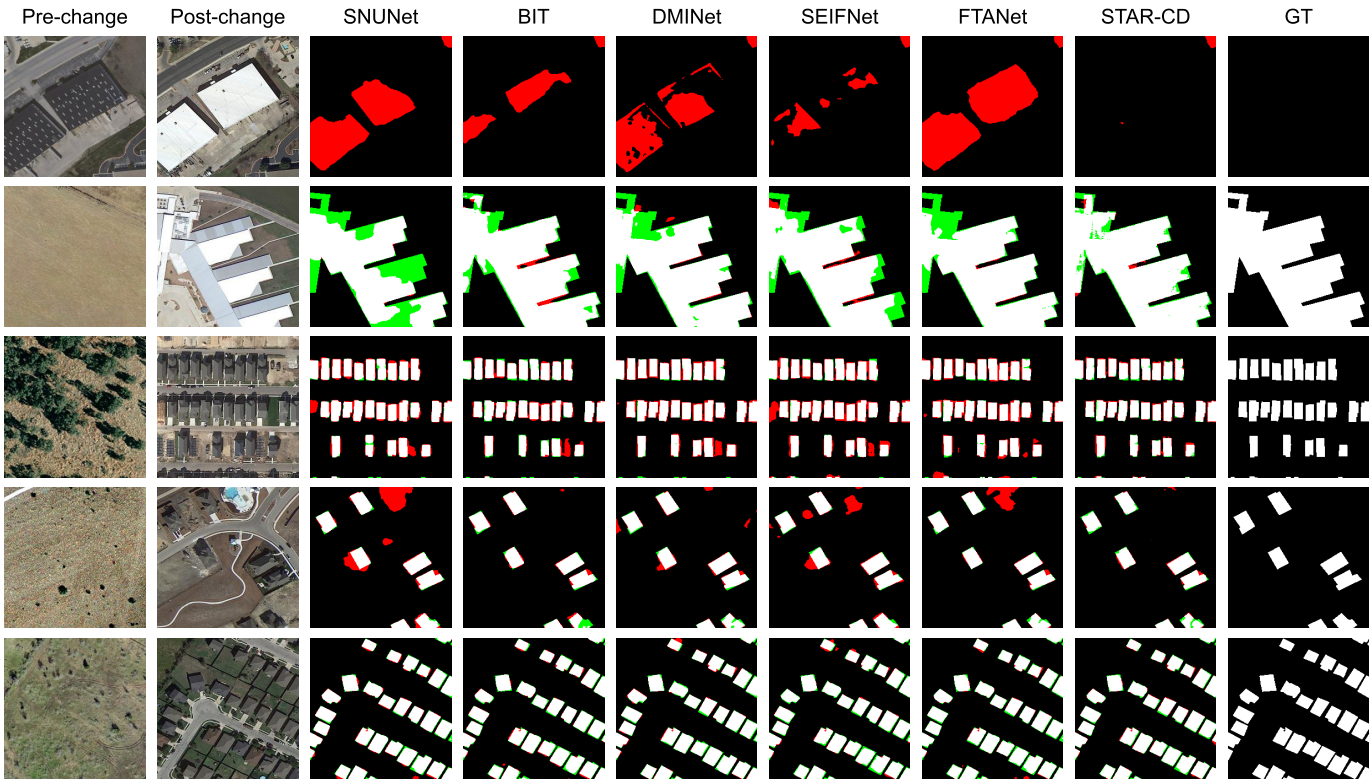


Fig. 7. Visual comparison result between our STAR-CD and the other state-of-the-art methods on the LEVIR-CD dataset. The true positives, true negatives, false positives, and false negatives are represented by white, black, red, and green, respectively.

predictions, which is caused by the lack of sufficient global relation change modeling, and it is addressed by our proposed ARMB. Through the above experimental analysis, we further validate the superiority of the proposed method.

D. Experimental Results on CDD Dataset

The CDD dataset [63] is derived from Google Earth (Digital Globe) and consists of seasonally varying remote sensing images of the same region. Unlike the WHU-CD and LEVIR-CD datasets, the CDD dataset is designed for general CD, covering a wide range of land-cover changes with an area larger than 500 pixels. This makes CDD more challenging. Following the official dataset split setting, the image numbers in the training, validation, and testing datasets are 10 000, 2998, and 3000, respectively.

The quantitative comparison results on the CDD dataset are reported in Table III. As evidenced by the results, our proposed STAR-CD demonstrates superior performance in general land cover CD, achieving remarkable improvements over existing state-of-the-art methods. In particular, STAR-CD consistently outperforms all competing approaches across every evaluation metric, with substantial advantages observed in three key metrics: $F1$, IoU, and OA. Specifically, compared to the second-best method SEIFNet, STAR-CD achieves significant gains of 2.40%, 4.34%, and 0.58% in $F1$ score, IoU, and OA, respectively, demonstrating its strong CD capability. Notably, the performance of current state-of-the-art methods is not consistent across different datasets. For example, although DMINet achieves promising results on WHU-CD and CDD, it falls behind FTANet on LEVIR-CD. In contrast, our proposed STAR-CD achieves the best detection performance on all datasets, demonstrating its robustness and generalization capability across various CD scenarios. Some visualization results on the CDD dataset are demonstrated in Fig. 8. Our proposed STAR-CD demonstrates significant advantages over existing methods in suppressing pseudo-change and accurately delineating the boundaries of changed regions. In the second and third scenes, some shadow regions are incorrectly identified as change areas by the SNUNet, DMINet, and SEIFNet, whereas

TABLE III
QUANTITATIVE RESULT ON THE CDD DATASET. BOLD REPRESENTS THE BEST RESULT, AND UNDERLINED REPRESENTS THE SECOND BEST. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Methods	Pre	Rec	OA	F1	IoU
FC-EF [13]	89.91	83.28	96.79	86.47	76.17
FC-Siam-Conc [13]	92.65	85.84	97.41	89.12	80.37
FC-Siam-Diff [13]	92.24	85.72	97.34	88.86	79.95
SNUNet [15]	93.57	89.72	97.97	91.60	84.51
BIT [24]	93.32	90.12	97.98	91.68	84.65
DMINet [16]	<u>94.76</u>	91.65	98.34	93.18	87.30
SEIFNet [17]	94.45	<u>92.29</u>	<u>98.38</u>	<u>93.36</u>	<u>87.53</u>
FTANet [21]	92.59	89.49	97.82	91.01	83.51
STAR-CD	**96.28**	**95.25**	**98.96**	**95.76**	**91.87**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

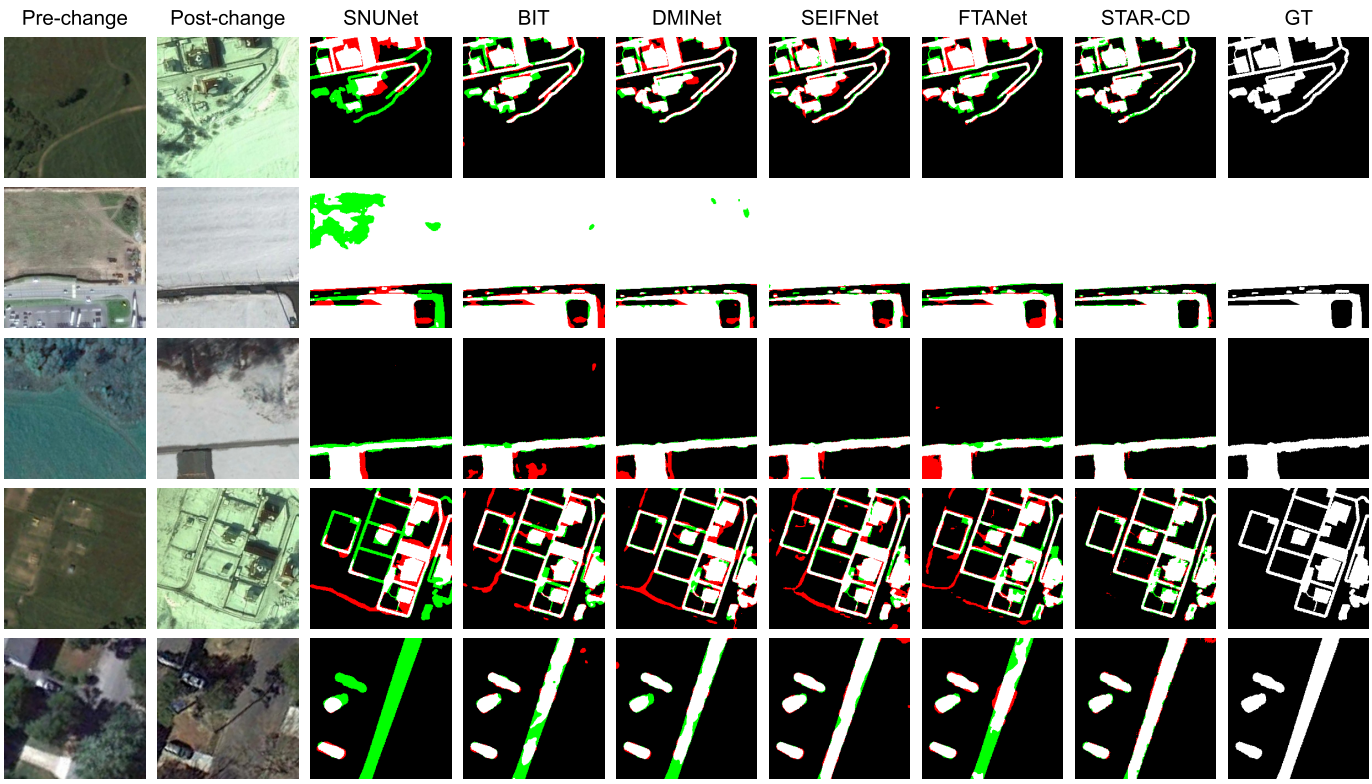


Fig. 8. Visual comparison result between our STAR-CD and the other state-of-the-art methods on the CDD dataset. The true positives, true negatives, false positives, and false negatives are represented by white, black, red, and green, respectively.

TABLE IV
ABLATION STUDY ON EACH COMPONENT IN STAR-CD. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Baseline	ARMB	FSAM	F1	IoU
✓			92.85	86.65
✓	✓		93.88	88.48
✓		✓	93.70	88.13
✓	✓	✓	**94.21**	**89.05**

TABLE V
ABLATION STUDY ON THE PROPOSED ARMB. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Baseline	Appearance	Relation	F1	IoU
✓			92.85	86.65
✓	✓		93.63	88.01
✓		✓	93.72	88.18
✓	✓	✓	**93.88**	**88.48**

our STAR-CD accurately distinguishes these pseudo-changes. In the fourth scene, only STAR-CD accurately identifies the changed regions, while all other methods mistakenly classify road areas with significant style differences as change regions. The above quantitative and qualitative results highlight the superiority of STAR-CD in general CD scenarios.

E. Ablation Study

1) Break-Down Ablation: To evaluate the contribution of each component in STAR-CD, we conduct a detailed ablation study. We report the $F1$ score of IoU of different variant models on the WHU-CD dataset. First, we establish a baseline model, which excludes the proposed ARMB and FSAM. As reported in Table IV, the baseline model achieves 92.85% points in $F1$ score and 86.65% points in IoU. Introducing ARMB would enhance the model's ability to capture differences from both local appearance and global relation perspectives, leading to an increase of 1.03% points in $F1$ and 1.83% points in IoU. For the FSAM, it helps surpass the style-related change noise and boost the detection performance by 0.85% points in $F1$ and 1.48% points in IoU. These two modules work collaboratively and bring 1.36% points $F1$ improvement and 2.40% points IoU improvement to the baseline model, demonstrating their effectiveness.

2) Ablation on Appearance-Relation Modeling: In the proposed ARMB, a dual-branch structure is adopted to simultaneously capture local appearance differences and global relational variations. In order to verify the effectiveness of ARMB, we designed two variant models, retaining the appearance modeling branch (which includes only the window-based MSA) and the relation modeling branch (which incorporates the spatial shuffle operation) of ARMB, respectively, and report their CD results in Table V. From the results, we can see that introducing either the appearance or the relation modeling branch individually brings notable improvements over the baseline. Specifically, adding only the appearance branch boosts the $F1$ score from 92.95% to 93.63% and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

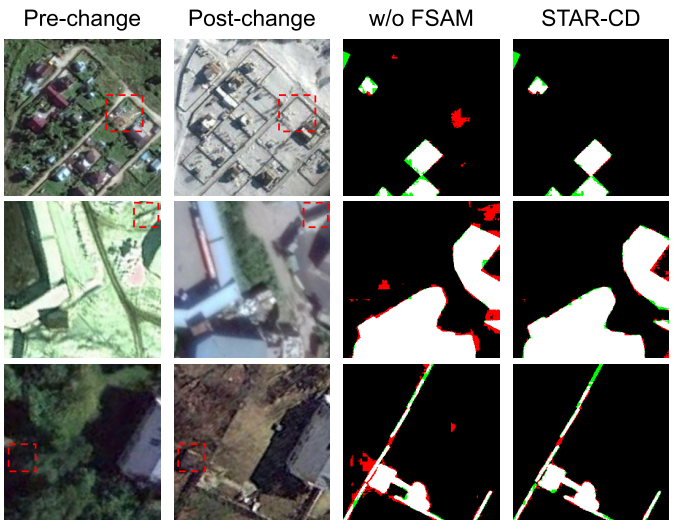


Fig. 9. Ablation study on the proposed FSAM, the true positives, true negatives, false positives, and false negatives are represented by white, black, red, and green, respectively. Our FSAM successfully suppressed style-semantic inconsistency.

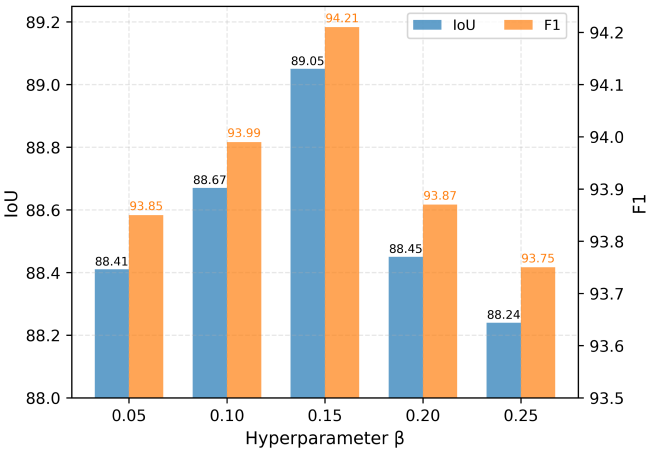


Fig. 10. Impact of the hyperparameter β in FSAM on CD performance. The F1 and IoU results on the WHU-CD dataset are reported.

IoU from 86.83% to 88.01%, demonstrating its effectiveness in capturing fine-grained texture differences. On the other hand, introducing only the relation modeling branch results in a slightly higher gain, improving the $F1$ to 93.72% and IoU to 88.18%, highlighting the importance of global context in reducing false alarms caused by isolated ambiguous patches. When both branches are integrated, the full ARMB achieves better performance, which confirms the complementarity of local appearance and global relational cues in enhancing overall CD accuracy.

3) Effectiveness of the Style Alignment: In our proposed STAR-CD, we incorporate an FSAM to mitigate the problem of style-semantic inconsistency. To validate its effectiveness, we conduct an ablation study on the CDD dataset and present the corresponding visual results in Fig. 9. As shown in the results, FSAM effectively suppresses style-induced change noise. Specifically, in scenes 1 and 3, introducing FSAM enables the model to ignore nonsemantic variations caused by seasonal vegetation growth. Similarly, in scene 2, FSAM helps the model avoid false detections caused by building shadows. These visual results demonstrate the effectiveness of FSAM in reducing style-related pseudo-changes.

4) Hyperparameter Analysis on the β in FSAM: In this section, we further perform sensitivity analysis on the hyperparameter β in the proposed FSAM to quantify its impact on the CD performance. Specifically, we vary $\beta \in [0.05, 0.25]$ and evaluate models on the WHU-CD dataset. As shown in Fig. 10, both IoU and $F1$ scores follow an inverse U trend within the range $\beta \in [0.05, 0.25]$, reaching their maximum at $\beta = 0.15$ with 89.05% in IoU and 94.21% in $F1$. When β is set to a small value, it fails to bridge the style discrepancies between bitemporal features, limiting FSAM's ability to suppress pseudo-changes. In contrast, a relatively large β tends to degrade performance, possibly due to the incorporation of high-frequency artifacts that distort structural information [54]. The above experiment results suggest that setting $\beta = 0.15$ achieves an optimal balance between effective style alignment and preservation of change-related structural integrity.

F. Robustness Evaluation Under Degraded Conditions

In real-world CD scenarios, data often exhibit high variability and less-controlled conditions due to various image degradations. To assess the robustness of different algorithms under such challenges, we further evaluate the performance of some representative methods by introducing Gaussian blur and Gaussian noise at different levels on the CDD-CD test set. Specifically, the standard deviation of Gaussian noise is set to three levels: 5, 10, and 15, while the Gaussian blur kernel standard deviation is set to 1.5, 2.0, and 2.5. To quantify robustness, we define a robustness index (RI) as follows:

$$\mathrm{RI} = \sum_{k=1}^{K} \frac{F1^{k}_{\mathrm{degra}}}{F1_{\mathrm{clean}}} \tag{27}$$

where K denotes the total number of degradation levels for each degradation type. A higher RI indicates stronger robustness against degradations. As shown in Fig. 11, our proposed STAR-CD demonstrates superior robustness against degradations compared with the four SOTA methods, especially under Gaussian noise degradation. The above evaluation results suggest that our proposed STAR-CD maintains robustness against data variability, implying a better potential to generalize to unseen or less controlled real-world scenarios where unpredictable degradations may occur.

G. Computational Complexity Analysis

In this section, we further consider the computational complexity and model size of different CD models. In Fig. 12, we demonstrated a detailed analysis of the model size and computational complexity of existing state-of-the-art methods. The computational cost of different methods is represented as the size of the scatter points, and is calculated when the input bitemporal remote sensing image size is $256 \times 256 \times 3 \times 2$, and $F1$ scores from the CDD dataset are reported. As can be seen from the results, the proposed STAR-CD achieves a considerable gain in the $F1$ score compared to other competitive

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

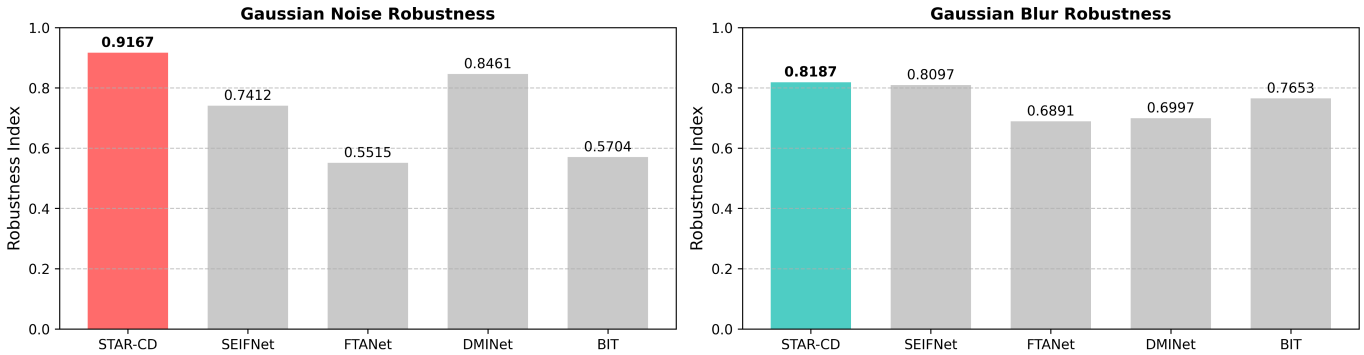


Fig. 11. Robustness comparison against common degradation factors of different methods.

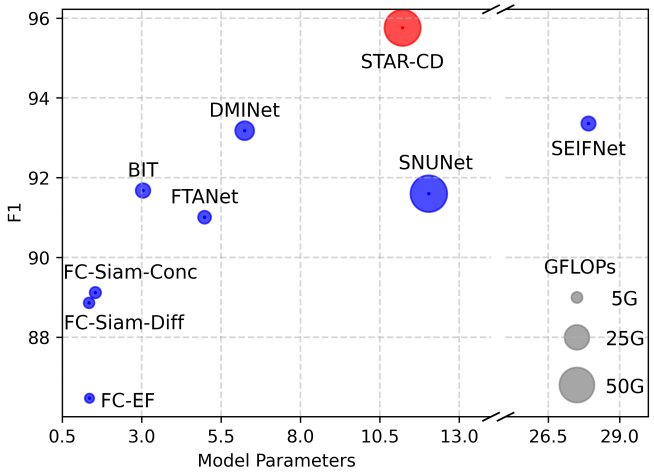


Fig. 12. $F1$, parameters, and FLOPs comparison with different CD methods. The $F1$ score is from the CDD dataset and is plotted on the vertical axis, while the model size is represented on the horizontal axis. The size of the scatter points reflects the computational complexity of the CD models.

CD methods, with comparable model sizes. STAR-CD also achieves a favorable performance-computation tradeoff. For example, while the model has a comparable number of parameters to SNUNet, it achieves an improvement of more than 4% in $F1$ score. More strikingly, STAR-CD uses less than half the parameters of SEIFNet, while still outperforming it by a considerable margin—over 2% in $F1$. These results underscore that our architecture achieves a superior performance-cost balance compared to existing methods. Nevertheless, STAR-CD has a relatively high computational complexity, which is mainly attributed to two factors. First, STAR-CD adopts a transformer-based backbone, which inherently requires more computation compared to conventional CNN-based CD models. Second, our proposed ARMB module introduces dual branches to separately capture local appearance variants and global relational differences. Although both branches still perform attention within local windows, this two-branch design inevitably increases the overall computational overhead.

H. Intermediate Feature Analysis

In this section, we investigate the characteristics of the proposed modules through intermediate feature analysis. Specifically, we visualize the heat activation maps of the bitemporal remote sensing images at the input and output of the ARMB. As shown in Fig. 13, since the ARMB is designed to model bitemporal feature interactions, the regions with high activations are not necessarily the final change areas. Nevertheless, it can be observed that the ARMB as a whole substantially enhances the feature responses around building regions in both pre- and post-change images, resulting in more comprehensive and salient representations. This effect can be attributed to the module's ability to jointly capture local and global contextual dependencies, which facilitates a more consistent and discriminative representation of potential change-relevant regions.

For the proposed FSAM, we visualize the absolute difference map between bitemporal features before and after being processed by FSAM, denoted as "DiffMap1" and "DiffMap2" in Fig. 14. Note that the two heatmaps of the same scene are displayed under the same colorbar for fair comparison. As shown in the results, before FSAM processing, the difference map exhibits only weak and scattered responses in the change regions, making it difficult to distinguish true changes from irrelevant variations. In contrast, after FSAM processing, the difference responses become significantly stronger and more spatially concentrated within the actual change areas. This observation suggests that FSAM effectively suppresses irrelevant feature discrepancies while emphasizing semantic variations between the bitemporal images, thereby facilitating a more reliable identification of real change regions.

V. Discussion and Future Work

While our proposed STAR-CD framework demonstrates significant improvements in capturing global change relationships and mitigating style-induced pseudo-changes, it still exhibits limitations in handling objects located near image boundaries. As illustrated in Fig. 15, both STAR-CD and several recent SOTA methods have difficulties accurately detecting or classifying changes involving objects that are cropped at the image borders. This limitation stems primarily from the insufficient semantic context of these incomplete targets, which hinders the model's ability to reliably discern the true change state or accurately classify the type of such fragmented objects. Therefore, achieving robust CD for incomplete objects remains a valuable research direction. Addressing this challenge would directly enhance the applicability of CD models in practical scenarios like urban expansion monitoring. Promising

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

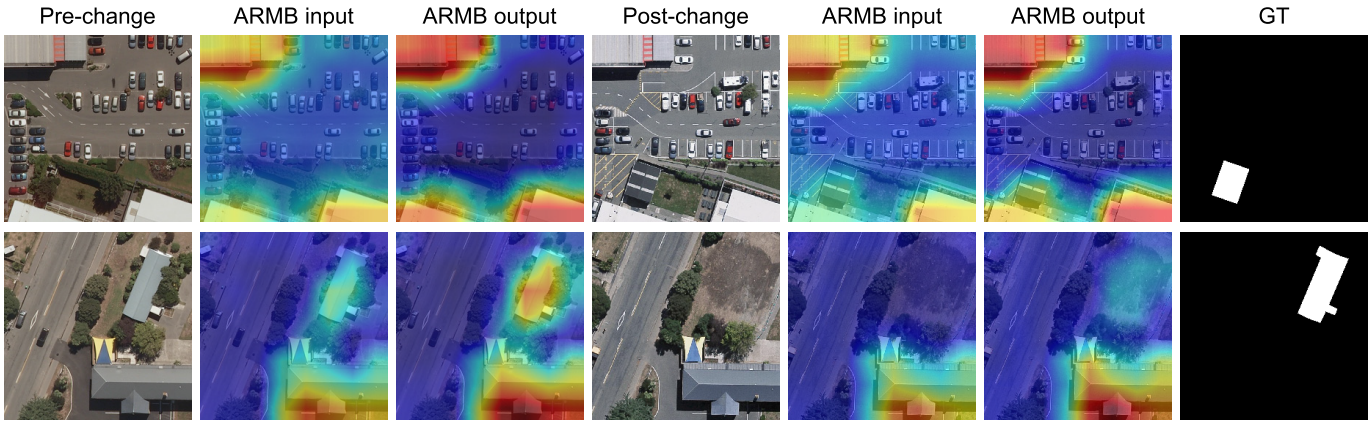


Fig. 13. Feature visualization results of the proposed ARMB, which enhances the representation of potential change-relevant regions.

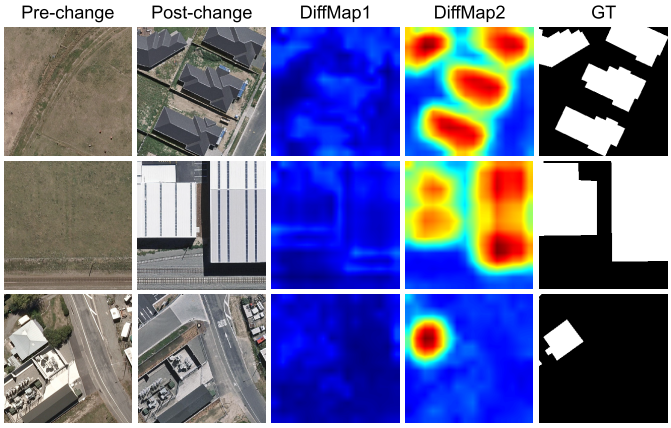


Fig. 14. Absolute difference maps of bitemporal features before and after being processed by FSAM, which suppresses irrelevant discrepancies while emphasizing semantic variations in changed regions.

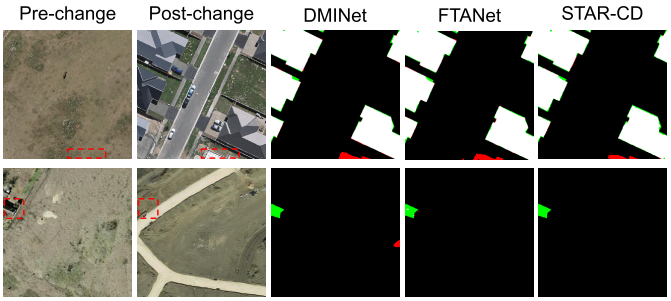


Fig. 15. Challenges in detecting changes for incomplete remote sensing objects at image boundaries. The true positives, true negatives, false positives, and false negatives are represented by white, black, red, and green, respectively.

future solutions may involve enhancing the model's semantic understanding of boundary regions or incorporating spatially adjacent patches during training to improve the model's comprehension of object spatial coherence.

Beyond this, there remain two other important open challenges to be explored in future work. First, data availability is often constrained in remote sensing CD. Annotating change labels not only requires twice the effort of standard segmentation tasks, but also necessitates annotators to have specialized expertise to differentiate between various ground objects. In addition, CD models typically exhibit poor generalization across different datasets due to domain shift, where each dataset possesses distinct ground features and change patterns. To alleviate this data scarcity challenge in CD, a promising approach is leveraging generative neural networks to synthesize large volumes of bitemporal remote sensing images along with their corresponding change labels, enhancing the generalizability of the models. Second, most existing methods rely on deterministic CD (changed/unchanged), failing to model the continuum of gradual changes prevalent in real-world scenarios. For instance, vegetation growth or degradation is inherently a progressive process, yet current frameworks struggle to quantify such intermediate states. While some approaches employ multiple discrete levels to assess building damage, such schemes remain discrete and are restricted to specific categories. A unified framework for general and continuous change state characterization is urgently needed to expand the applicability of CD technologies.

VI. Conclusion

In this article, we present an accurate bitemporal CD framework named STAR-CD, with two designed bitemporal feature interaction mechanisms to overcome the performance bottlenecks faced by existing CD models. We observe that insufficient modeling of global relations in remote sensing scenes often leads to coarse change boundaries and misclassification of change regions. To address this problem, we propose the appearance relation modeling block, which simultaneously explores change cues from local appearance differences and global relation variations. Furthermore, due to variations in illumination, atmospheric conditions, and imaging sensors, objects with identical semantics often exhibit significant style inconsistencies across bitemporal images. This cross-temporal semantic-style inconsistency can mislead CD models by introducing pseudo-changes. To mitigate this, we propose the FSAM, which decouples texture and style information in the frequency domain, thereby aligning cross-temporal style information and suppressing style-induced noise. To validate the effectiveness of our methods, we conduct extensive experiments on three benchmark datasets, WHU-CD, LEVIR-CD, and CDD. Experimental results demonstrate that STAR-CD consistently outperforms state-of-the-art methods across most evaluation metrics. In addition, qualitative visualization results

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

further highlight the superiority of our method in producing more accurate and finer-grained change predictions, while effectively overcoming the interference caused by style inconsistencies between bitemporal images.

REFERENCES

[1] R. E. Kennedy et al., "Remote sensing change detection tools for natural resource managers: Understanding concepts and tradeoffs in the design of landscape monitoring projects," *Remote Sens. Environ.*, vol. 113, no. 7, pp. 1382–1396, Jul. 2009.

[2] P. de Bem, O. de Carvalho Junior, R. Fontes Guimarães, and R. Trancoso Gomes, "Change detection of deforestation in the Brazilian Amazon using Landsat data and convolutional neural networks," *Remote Sens.*, vol. 12, no. 6, p. 901, Mar. 2020.

[3] M. Pirnazar et al., "Change detection of urban land use and urban expansion using GIS and RS, case study: Zanjan province, Iran," *Int. J. Constructive Res. Civil Eng.*, vol. 4, no. 1, pp. 295–309, 2018.

[4] E. Hawash, A. El-Hassanin, W. Amer, A. El-Nahry, and H. Effat, "Change detection and urban expansion of port sudan, red sea, using remote sensing and GIS," *Environ. Monitor. Assessment*, vol. 193, no. 11, pp. 1–22, Nov. 2021.

[5] F. Bovolo and L. Bruzzone, "A split-based approach to unsupervised change detection in large-size multitemporal images: Application to tsunami-damage assessment," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1658–1670, Jun. 2007.

[6] D. Qin et al., "MSIM: A change detection framework for damage assessment in natural disasters," *Expert Syst. Appl.*, vol. 97, pp. 372–383, May 2018.

[7] G. A. Olson, A. Cheriyadat, P. Mali, and C. G. O'Hara, "Detecting and managing change in spatial data-land use and infrastructure change analysis and detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2004, pp. 729–734.

[8] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and *k*-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[9] R. D. Johnson and E. S. Kasischke, "Change vector analysis: A technique for the multispectral monitoring of land cover and condition," *Int. J. Remote Sens.*, vol. 19, no. 3, pp. 411–426, Jan. 1998.

[10] Y. V. Vizilter, A. Rubis, S. Zheltov, and O. Vygolov, "Change detection via morphological comparative filters," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 279–286, Jun. 2016.

[11] B. Hou, Y. Wang, and Q. Liu, "Change detection based on deep features and low rank," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2418–2422, Dec. 2017.

[12] M. Gong, P. Zhang, L. Su, and J. Liu, "Coupled dictionary learning for change detection from multisource data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7077–7091, Dec. 2016.

[13] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[14] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[15] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[16] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[17] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.

[18] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[19] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[20] H. Chang, P. Wang, W. Diao, G. Xu, and X. Sun, "Remote sensing change detection with bitemporal and differential feature interactive perception," *IEEE Trans. Image Process.*, vol. 33, pp. 4543–4555, 2024.

[21] T. Zhu et al., "FTA-Net: Frequency-temporal-aware network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3448–3460, 2025.

[22] J. Meng et al., "ChangeDA: Depth-augmented multitask network for remote sensing change detection via differential analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5616119.

[23] Z. Li, C. Tang, L. Wang, and A. Y. Zomaya, "Remote sensing change detection via temporal feature interaction and guided refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628711.

[24] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[25] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[26] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.

[27] S. Marchesi and L. Bruzzone, "ICA and kernel ICA for change detection in multispectral remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2009, pp. II-980–II-983.

[28] G. Chen, G. J. Hay, L. M. T. d. Carvalho, and M. A. Wulder, "Object-based change detection," *Int. J. Remote Sens.*, vol. 33, no. 14, pp. 4434–4457, 2012.

[29] P. Xiao, X. Zhang, D. Wang, M. Yuan, X. Feng, and M. Kelly, "Change detection of built-up land: A framework of combining pixel-based detection and object-based recognition," *ISPRS J. Photogramm. Remote Sens.*, vol. 119, pp. 402–414, Sep. 2016.

[30] L. Xu, W. Jing, H. Song, and G. Chen, "High-resolution remote sensing image change detection combined with pixel-level and object-level," *IEEE Access*, vol. 7, pp. 78909–78918, 2019.

[31] Y. Yuan, X. Chen, K. Tang, and J. Chen, "A 'difference-in-differences'-based method for unsupervised change detection in season-varying images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[32] J. Miao, S. Li, X. Bai, W. Gan, J. Wu, and X. Li, "RS-NormGAN: Enhancing change detection of multi-temporal optical remote sensing images through effective radiometric normalization," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 324–346, Mar. 2025.

[33] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[34] A. Codegoni, G. Lombardi, and A. Ferrari, "TINYCD: A (not so) deep learning model for change detection," *Neural Comput. Appl.*, vol. 35, no. 11, pp. 8471–8486, Apr. 2023.

[35] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.

[36] Y. Liu, F. Liu, J. Liu, and L. Xiao, "Difference guidance learning with feature alignment for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635015.

[37] F. Zhou, X. Zhang, H. Shuai, R. Hang, S. Zhu, and T. Geng, "Dual-granularity feature alignment for change detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 4487–4497, 2025.

[38] W. Cui et al., "Multi-scale semantic segmentation and spatial relationship recognition of remote sensing images based on an attention model," *Remote Sens.*, vol. 11, no. 9, p. 1044, May 2019.

[39] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[40] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4096–4105.

[41] Y. Zhang, H. Song, Q. Wang, P. Jin, and T. Shen, "Semantic co-occurrence and relationship modeling for remote sensing image segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 6630–6640, 2025.

[42] Z. Tian, R. Zhan, W. Wang, Z. He, J. Zhang, and Z. Zhuang, "Object detection in optical remote sensing images by integrating object-to-object relationships," *Remote Sens. Lett.*, vol. 11, no. 5, pp. 416–425, May 2020.

[43] S. Tian, L. Kang, X. Xing, J. Tian, C. Fan, and Y. Zhang, "A relation-augmented embedded graph attention network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 1000718.

[44] R. Cong, Y. Zhang, L. Fang, J. Li, Y. Zhao, and S. Kwong, "RRNet: Relational reasoning network with parallel multiscale attention for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613311.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.

[45] G. Liu, Y. Yuan, Y. Zhang, Y. Dong, and X. Li, "Style transformation-based spatial–spectral feature learning for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5401515.
[46] Z. Zhang, C. Liu, F. Hao, and Z. Liu, "Style-transfer-based unsupervised change detection from heterogeneous images," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 61, no. 3, pp. 6537–6550, Jun. 2025.
[47] Z.-G. Liu, Z.-W. Zhang, Q. Pan, and L.-B. Ning, "Unsupervised change detection from heterogeneous data based on image translation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403413.
[48] Q. Han et al., "StyleFormer: Spatial–temporal style projecting bidirectional interactive transformer for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5609016.
[49] S. Fu, S. Dong, and X. Meng, "Beyond cross-temporal difference: style-aligned and fusion-difference learning for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631415.
[50] Z. Yang, X. Wang, M. Li, and J. Long, "SCEDNet: A style consistency enhanced differential network for remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.
[51] Z. Huang, Y. Ben, G. Luo, P. Cheng, G. Yu, and B. Fu, "Shuffle transformer: Rethinking spatial shuffle for vision transformer," 2021, *arXiv:2106.03650*.
[52] Y. Cai et al., "Degradation-aware unfolding half-shuffle transformer for spectral compressive imaging," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 37749–37761.
[53] S.-Q. Deng, L.-J. Deng, X. Wu, R. Ran, D. Hong, and G. Vivone, "PSRT: Pyramid shuffle-and-reshuffle transformer for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503715.
[54] Y. Yang and S. Soatto, "FDA: Fourier domain adaptation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4084–4094.
[55] S. Lee, J. Bae, and H. Y. Kim, "Decompose, adjust, compose: Effective normalization by playing with frequency for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 11776–11785.
[56] H. Niu, L. Xie, J. Lin, and S. Zhang, "Exploring semantic consistency and style diversity for domain generalized semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 6, pp. 6245–6253.
[57] Q. Xu, R. Zhang, Y. Zhang, Y. Wang, and Q. Tian, "A Fourier-based framework for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14383–14392.
[58] M. Yi-de, L. Qing, and Q. Zhi-Bai, "Automated image segmentation using improved PCNN model based on cross-entropy," in *Proc. Int. Symp. Intell. Multimedia, Video Speech Process.*, Oct. 2004, pp. 743–746.
[59] C. H. Sudre, W. Li, T. Vercauteren, S. Ourselin, and M. J. Cardoso, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, Sep. 2017, pp. 240–248.
[60] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Álvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 12077–12090.
[61] H. Hu et al., "Learning implicit feature alignment function for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 487–505.
[62] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[63] M. Lebedev, Y. V. Vizilter, O. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *The Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 565–571, May 2018.



Qian Hu received the B.S. and M.S. degrees from the Electronic Information School, Wuhan University, Wuhan, China, in 2022 and 2025, respectively, where he is currently pursuing the Ph.D. degree.

His research interests include computer vision and image processing.



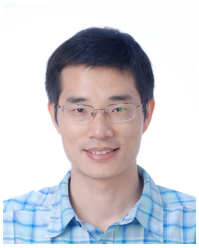
Yuxuan Wang received the B.S. degree in measurement and control technology and instrumentation from Wuhan University, Wuhan, China, in 2021, where he is currently pursuing the Ph.D. degree in communication and information systems.

His research involves computer vision and image processing, especially in cross-modality learning, semantic segmentation, and instance segmentation.



Yan Zhang received the B.S. degree in communication engineering from Wuhan University, Wuhan, China, in 2022, where she is currently pursuing the Ph.D. degree in computer science and technology.

Her research involves computer vision, especially in multimodal object detection and tiny object detection.



Wen Yang (Senior Member, IEEE) received the B.S. degree in electronic apparatus and surveying technology, the M.S. degree in computer application technology, and the Ph.D. degree in communication and information systems from Wuhan University, Wuhan, China, in 1998, 2001, and 2004, respectively.

From 2008 to 2009, he was a Visiting Scholar with the Apprentissage et Interfaces (AI) Team, Laboratoire Jean Kuntzmann, Grenoble, France. From 2010 to 2013, he was a Post-Doctoral Researcher with the State Key Laboratory of Information Engineering, Surveying, Mapping and Remote Sensing, Wuhan University. Since then, he has been a Full Professor with the School of Electronic Information, Wuhan University. His research interests include object detection and recognition, multisensor information fusion, and remote sensing image processing.



Fang Xu received the B.S. degree in electronic and information engineering and the Ph.D. degree in communication and information systems from Wuhan University, Wuhan, China, in 2018 and 2023, respectively.

She is currently a Post-Doctoral Researcher with the School of Computer Science, Wuhan University. Her research involves remote sensing image processing, including multimodal data matching and fusion.



Gui-Song Xia (Senior Member, IEEE) received the Ph.D. degree in image processing and computer vision from CNRS LTCI, Télécom ParisTech, Paris, France, in 2011.

From 2011 to 2012, he was a Post-Doctoral Researcher with the Center de Recherche en Mathématiques de la Décision, CNRS, Paris Dauphine University, Paris, for one and a half years. He was a Visiting Scholar with DMA, École Normale Supérieure (ENS-Paris), Paris, for two months in 2018. He is currently a Full Professor of computer vision and photogrammetry with Wuhan University, Wuhan, China. His research interests include mathematical modeling of images and videos, structure from motion, perceptual grouping, and remote sensing image understanding.

Dr. Xia serves on the editorial boards for several journals, including *ISPRS Journal of Photogrammetry and Remote Sensing*, *Pattern Recognition*, *Signal Processing: Image Communication*, *EURASIP Journal on Image and Video Processing*, *Journal of Remote Sensing*, and *Frontiers in Computer Science: Computer Vision*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 15:13:45 UTC from IEEE Xplore. Restrictions apply.